

MODULE ASSIGNMENT PYTHON 5

MARKETING CAMPAIGN PERFORMANCE INSIGHTS

Executive Summary

In the digital marketing environment, continuous evaluation of campaign performance is mandatory to maximize return on investment (ROI), minimize acquisition costs, and boost conversion rates. This report provides a structured, non-technical analysis of marketing campaign performance across diverse channels, audience segments, campaign types, and geographical regions.

Descriptive Analysis & Dataset Overview

Dataset Structure & Integrity

- Data Profiling:** The analysis evaluates dataset shape, column formats, memory consumption, and non-null values to ensure seamless analytical processing.
- Numerical Summary:** Descriptive statistics (Mean, Standard Deviation, Median, Min, and Max) are computed for key numerical metrics including ROI, Acquisition Cost, Conversion Rate, Clicks, and Impressions.

```
Descriptive Analysis
print(df.head())

   Campaign_ID  Company  Campaign_Type  Target_Audience  Duration  \
0             1         Tech          Influencer        Women 20-30  30 Days
1             2         Health          Influencer        Men 35-45  45 Days
2             3         Beauty          Social Media        Women 20-30  60 Days
3             4         Finance          Influencer        Men 35-45  45 Days
4             5         Tech          Influencer        Men 20-30  30 Days

Channel Used  Conversion_Rate  Acquisition_Cost  ROI  Location  \
0 Facebook          1.200394           5444    62.94    Houston
1 Google Ads          1.120375           1793    34.47  Washington, D.C.
2 Instagram          4.462375           5111    73.28    Miami
3 YouTube            4.464375           7820    68.92    Seattle
4 Google Ads          4.400538           2346    135.82    Chicago

Language  Clicks  Impressions  Engagement_Score  Customer_Segment  \
0 English    9801          67926              5      Tech Enthusiasts
1 German    1245          66363              4      Foodies
2 Spanish   1024          80248              8      Fashionistas
3 Chinese   1285          59251              6      Fitness
4 English   9870          94907              5      Tech Enthusiasts

Date
0  01-01-2023
1  01-01-2023
2  01-01-2023
3  01-01-2023
4  01-01-2023
```

```
Total Rows and Columns
print("Rows and Columns:", df.shape)

Rows and Columns: (22029, 16)

# Data Types and Non-null values
df.info()
<class 'pandas.core.frame.DataFrame'>
Int64Index: 22029 entries, 0 to 22028
Data columns (total 16 columns):
#   Column                Non-Null Count  Dtype  \
0   Campaign_ID           22029 non-null  int64   \
1   Company               22029 non-null  object  \
2   Campaign_Type         22029 non-null  object  \
3   Target_Audience      22029 non-null  object  \
4   Duration              22029 non-null  object  \
5   Channel_Used          22029 non-null  object  \
6   Conversion_Rate       22029 non-null  float64 \
7   Acquisition_Cost      22029 non-null  int64   \
8   ROI                  22029 non-null  float64 \
9   Location              22029 non-null  object  \
10  Language              22029 non-null  object  \
11  Clicks                22029 non-null  int64   \
12  Impressions           22029 non-null  int64   \
13  Engagement_Score      22029 non-null  int64   \
14  Customer_Segment      22029 non-null  object  \
15  Date                  22029 non-null  object  \
dtypes: float64(2), int64(5), object(9)
memory usage: 2.7+ MB
```

```
# Numerical columns
print(df.describe())

   Campaign_ID  Conversion_Rate  Acquisition_Cost  ROI  \
count  22029.000000    22029.000000    22029.000000  22029.000000
mean    11015.000000         4.757212         5522.748842    112.861648
std     6359.368876         0.966893        2597.666268     301.619721
min         1.000000         1.000000         1000.000000     -58.300000
25%    11015.000000         4.761527        5525.000000     93.650000
75%    10522.000000         5.429325        7766.000000    247.310000
max    22029.000000        7.469987        9999.000000   3189.750000

   Clicks  Impressions  Engagement_Score  \
count  22029.000000    22029.000000    22029.000000
mean    2223.867572    58618.460787     6.582323
std    1394.166330    25542.979123     1.450084
min         30.000000    1901.000000     4.000000
25%    1867.000000    25894.000000     5.000000
75%    2081.000000    58051.000000     7.000000
max    6887.000000   99999.000000     9.000000
```

Categorical Data Exploration

- Campaign Identification:** Distinct count of Campaign IDs is recorded to confirm individual campaign tracking integrity.
- Category Frequencies:** Identifies unique geographical locations and customer segment profiles while summarizing distribution counts across campaign types and promotion channels.

```
Category Frequencies
print(df.groupby('Category').size())

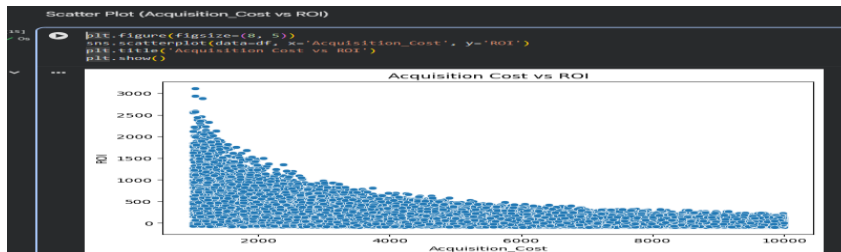
Category
Campaign_Type    5
Company           5
Customer_Segment  5
Location          5
Language          5
Target_Audience  5
Channel_Used      5
Date              5
```

Exploratory Data Analysis & Strategic Insights

Campaign Performance Dynamics

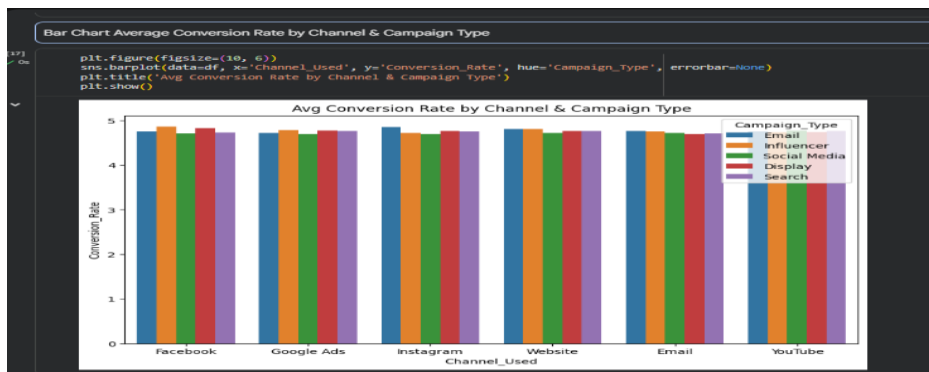
Scatter Plot (Acquisition Cost vs. ROI)

Examines the relationship between customer acquisition expenditure and overall campaign profitability to evaluate spending efficiency.



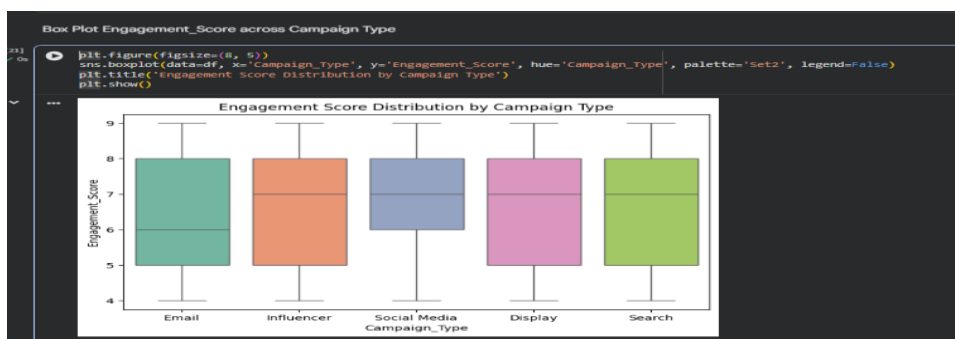
BAR CHART (Channel & Type Conversion Rate)

Identifies which marketing channels (e.g., Google Ads, Social Media) yield the highest average conversion rate when paired with specific campaign types.



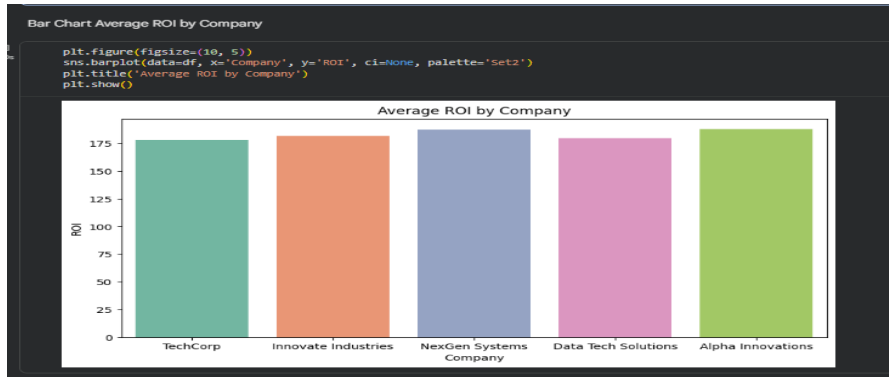
BOX PLOT (Engagement Score Distribution)

Analyzes customer interaction scores (1 to 10) across campaign categories to identify formats that consistently drive higher user interest.



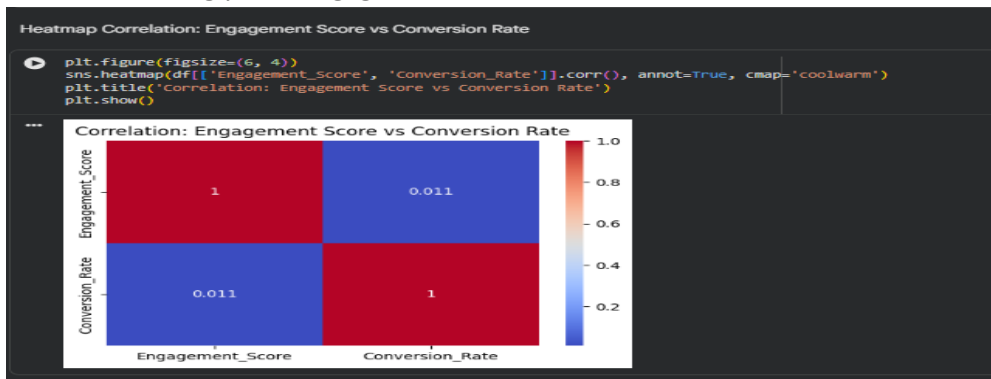
BAR CHART (Company Profitability Comparison)

Compares average ROI across operating companies/brands to benchmark relative marketing success.



HEATMAP (Engagement vs. Conversion Correlation)

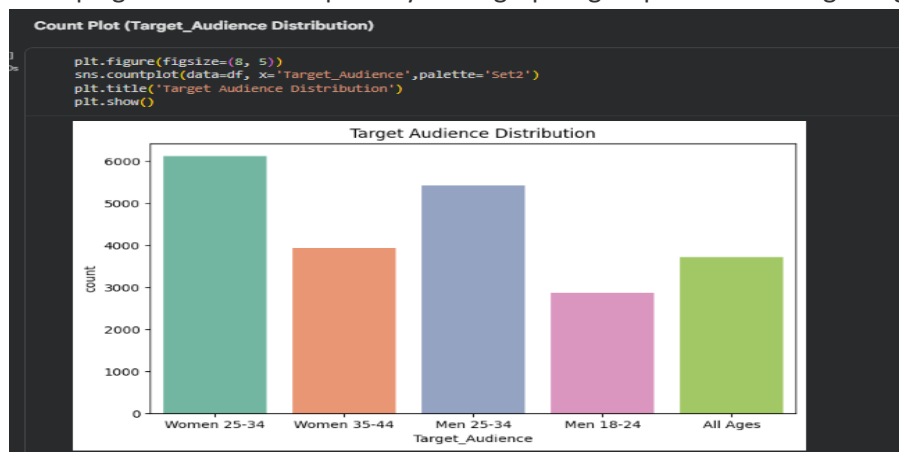
Measures how strongly user engagement score correlates with actual conversion rates.



Customer Segmentation & Localization

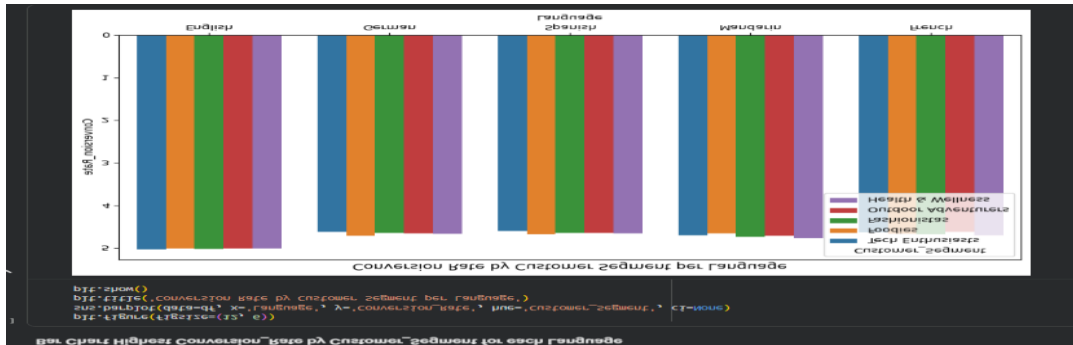
COUNT PLOT (Target Audience Reach)

Assesses campaign volume across primary demographic groups to reveal target segment focus.



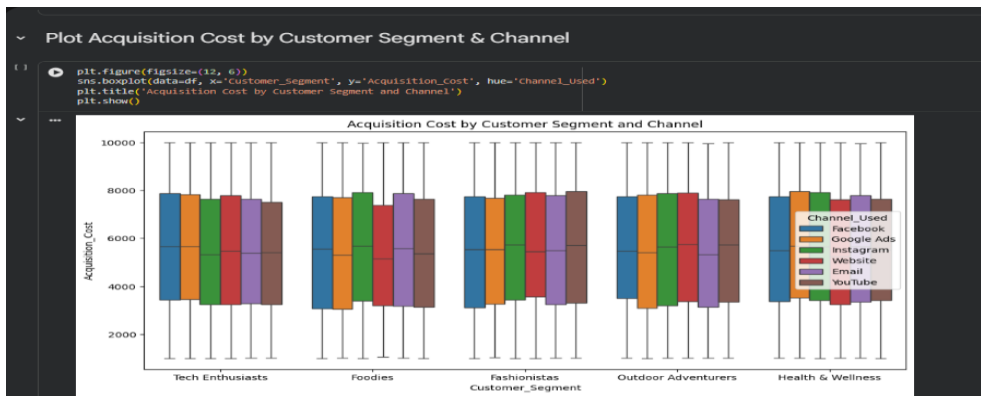
BAR CHART (Segment Efficiency by Language)

Pinpoints customer segments achieving peak conversion rates across distinct content languages.



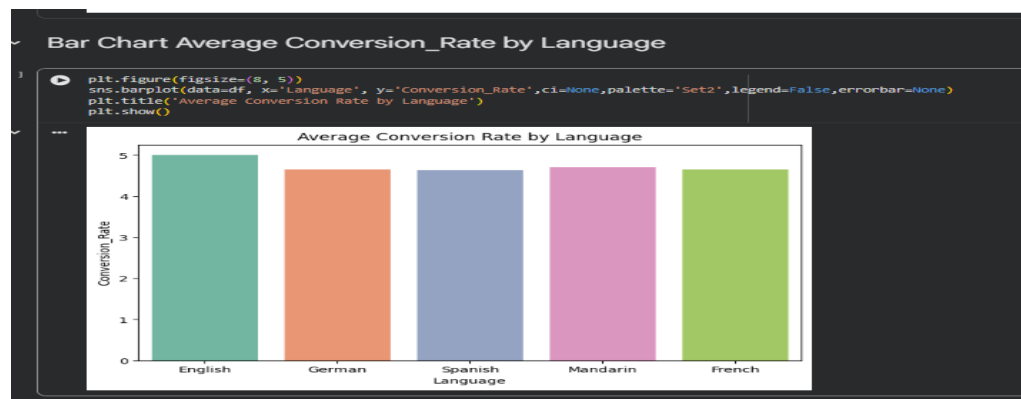
PLOT CHART (Acquisition Expense Spread)

Evaluates variations in customer acquisition expenses per segment across selected marketing channels.



BAR CHART (Language Effectiveness)

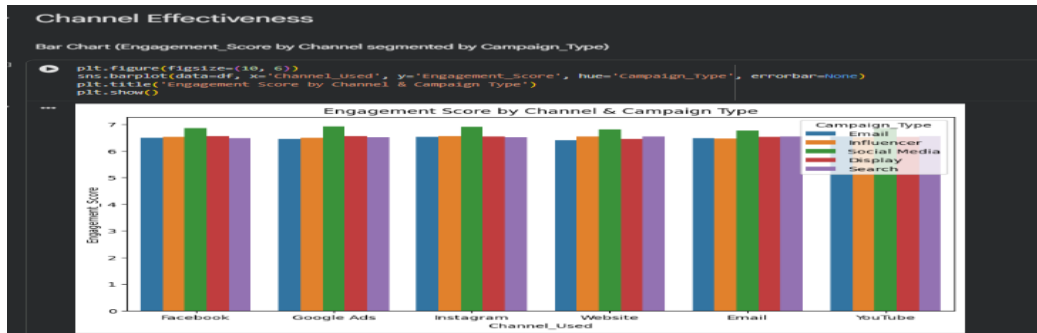
Compares aggregate conversion rates by language to optimize localized ad messaging.



Channel Effectiveness & Attribution

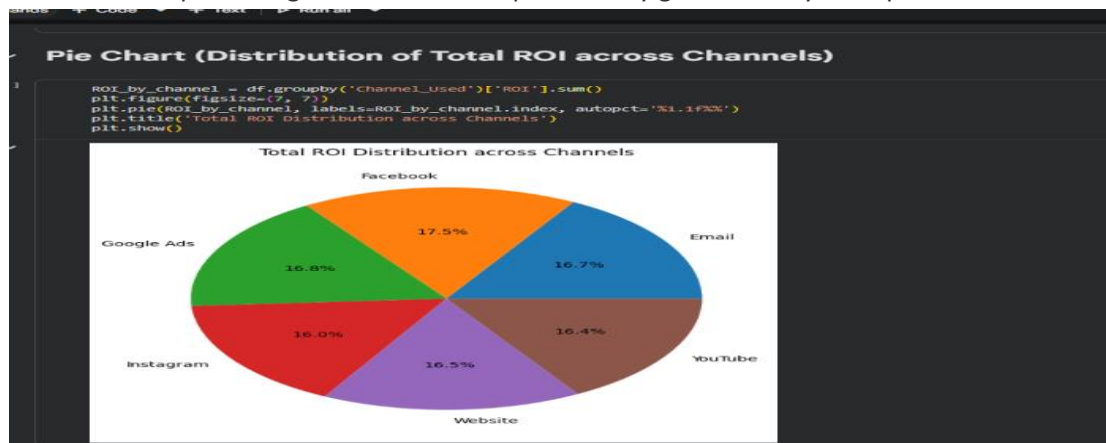
BARChart (Channel Engagement Levels)

Highlights top-performing interaction platforms across email, display, social media, and search campaigns.



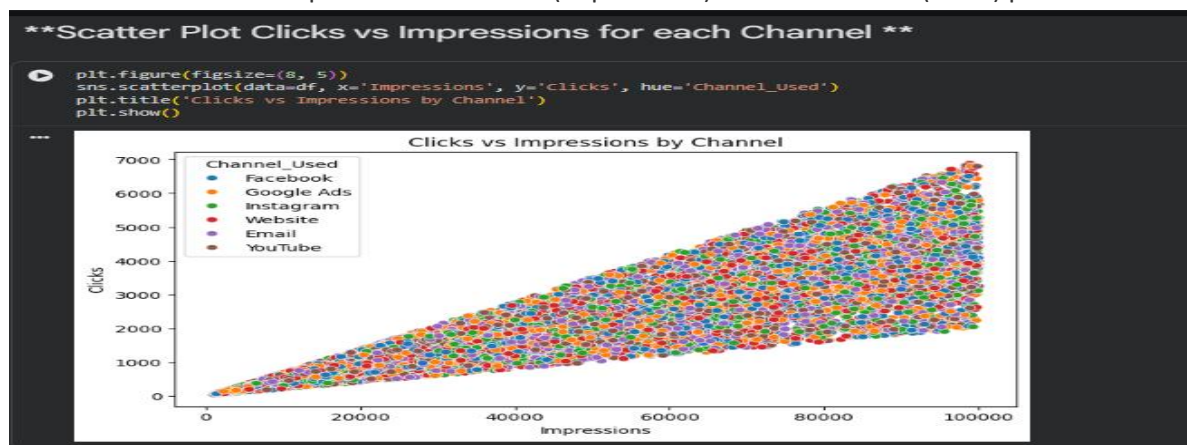
PIE CHART (Cumulative ROI Share)

Illustrates the percentage share of overall profitability generated by each promotion medium.



SCATTER PLOT CHART (Clicks vs. Impressions (CTR Potential))

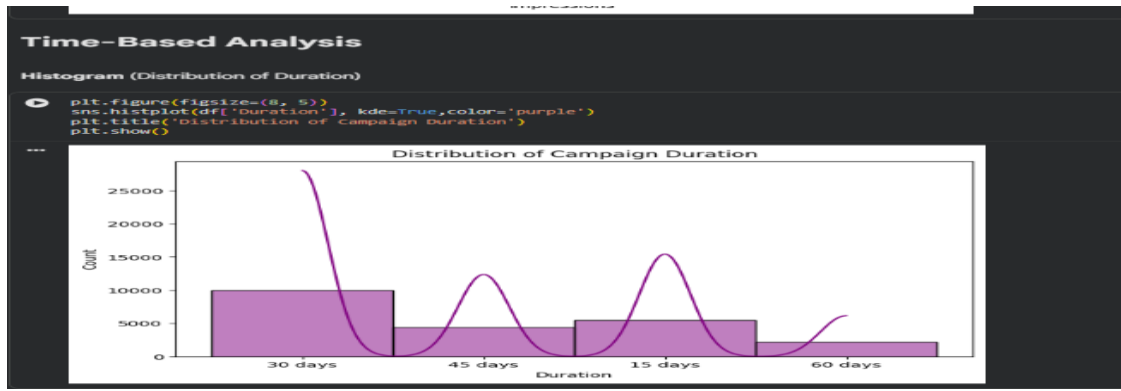
Examines the relationship between ad views (impressions) and user actions (clicks) per channel.



Time-Based Performance

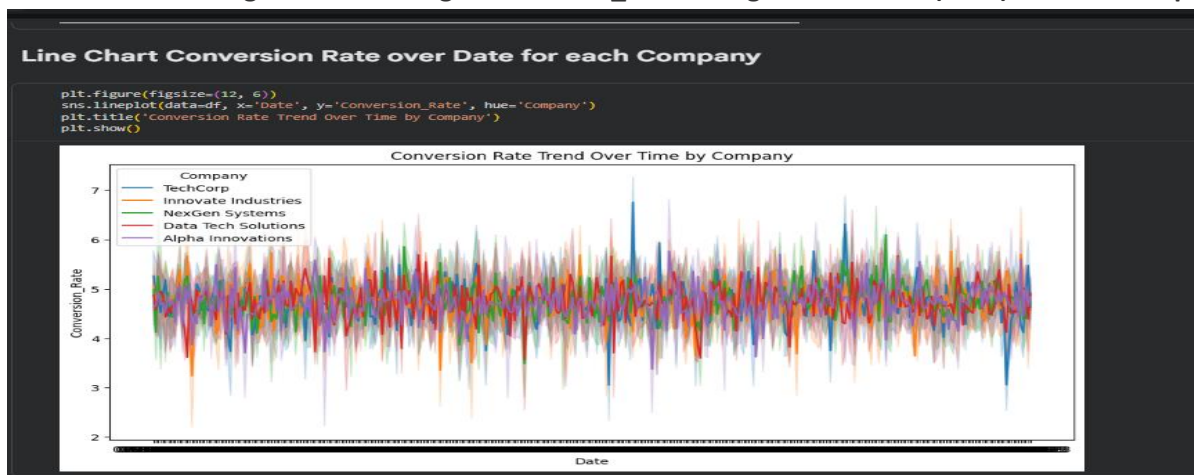
HISTOGRAM CHART (Campaign Duration Distribution)

A histogram showing the distribution of campaign lengths (Duration in days).



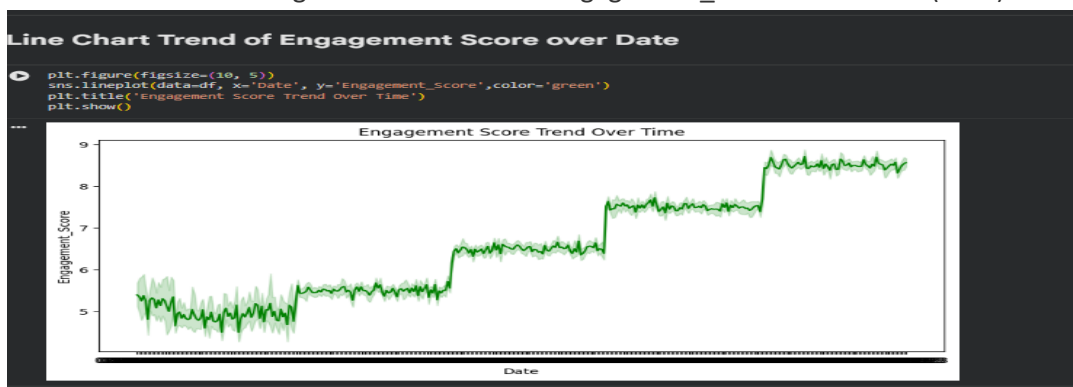
LINE CHART (Company Conversion Rate Over Time)

A line chart tracking how the average Conversion_Rate changed over dates (Date) for each company.



LINE CHART (Engagement Score Trend Over Time)

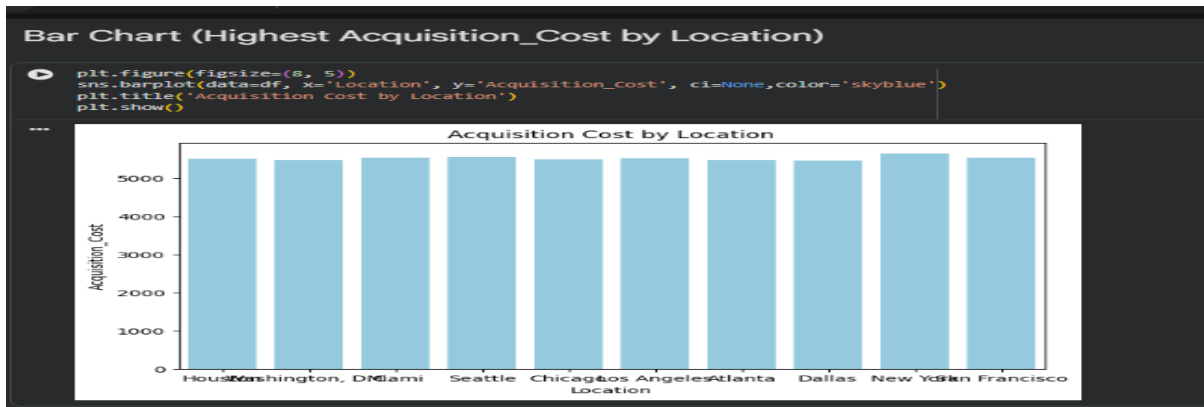
A line chart showing the overall trend of Engagement_Score across time (Date).



Temporal & Geographical Trends

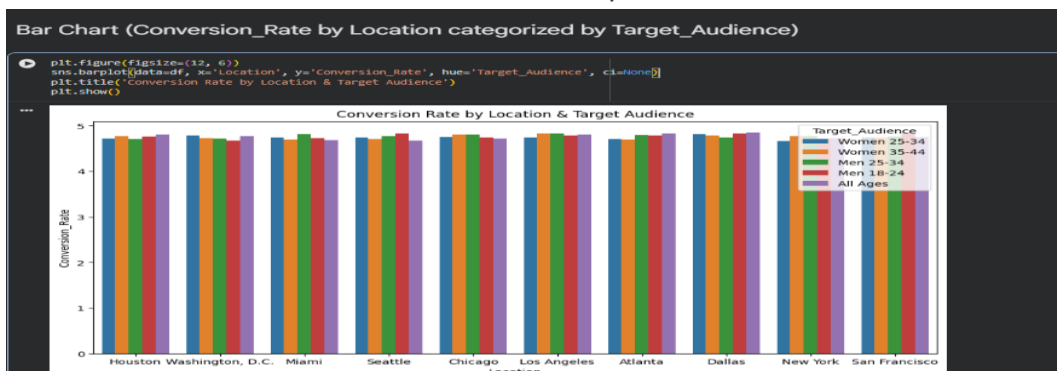
BAR CHART (Campaign Duration Distribution)

Evaluates optimal campaign run lengths (in days) to prevent audience fatigue.



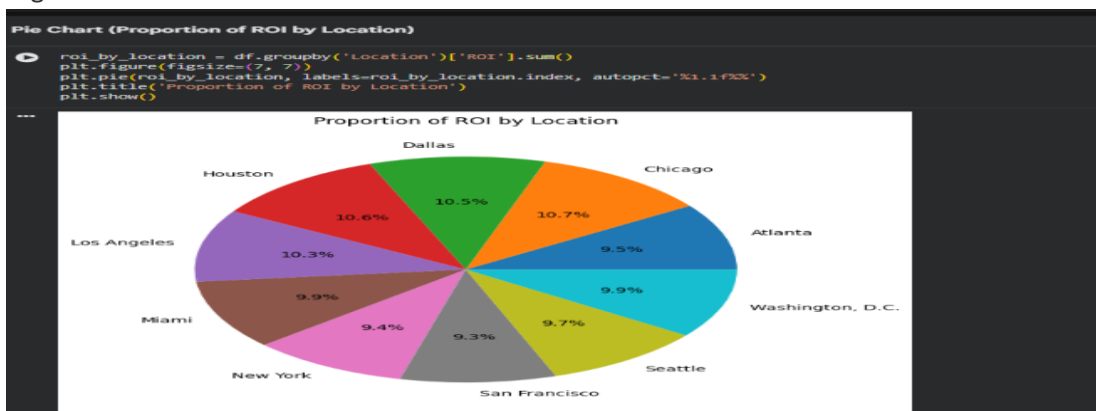
BAR CHART (Time-Series Conversion Dynamics)

Monitors conversion rate fluctuations over time by brand to detect seasonal trends.



PIE CHART (Geographical Cost & ROI Profiling)

Maps acquisition expenses and total profitability across operating locations to prioritize high-yield regions.



Conclusion & Strategic Recommendations

Key Conclusions

ROI & Channel Optimization:

The analysis proves that higher acquisition costs do not automatically guarantee higher ROI. Marketing budgets should be reallocated toward channels delivering consistently high Return on Investment rather than high expense platforms.

Target Demographic Alignment:

Conversion efficiency varies significantly across localized languages and target customer segments. Tailoring campaign content to regional language preferences yields noticeably higher conversion rates.

Engagement-to-Conversion Impact:

Higher engagement scores show a positive correlation with conversion success, confirming that interactive and highly engaging campaign formats drive meaningful bottom-line outcomes.

Actionable Recommendations

- **1. Budget Reallocation:** Shift capital away from underperforming channels toward high-ROI combinations identified in the channel effectiveness analysis.
- **2. Personalized Localization:** Expand multi-lingual campaign variations for top-performing customer segments in high-converting regions.
- **3. Optimized Duration:** Structure future campaigns according to optimal duration ranges to maintain maximum user engagement without overspending.